



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
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Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic –1.12 Static Class Members and Inline Function

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Static Class Members

Two types of static class members are:

- Static Data Members
- Static Member Functions



STATIC DATA MEMBERS

- We can define class members static using static keyword.
- A static member variable has certain special characteristics:
 - i. Only one copy shared by all objects of the class.
 - ii. Initialized to zero when the first object is created.
 - iii. Initialized outside the class.



STATIC MEMBER FUNCTIONS

- A static functions can access only other static members.
- A static function is called using class name and scope resolution operator.

Class-name :: function-name;

- It can not use this pointer.



EXAMPLE

```
#include <iostream>
using namespace std;
class test
{
    int code;
    static int count;
public:
    void set()
    {
        code=++count;
    }
    void show()
    {
        cout<<"Object number: "<<code<<endl;
    }
    static void showcount( )
    {
        cout<<"Count:"<<count<<endl;
    }
};
```

```
};
int test::count;
int main()
{
    test t1,t2,t3;
    t1.set();
    test::showcount();
    t2.set();
    test::showcount();
    t3.set();
    test::showcount();
    t1.show();
    t2.show();
    t3.show();
    return 0;
}
```

```
Count : 1
Count : 2
Count : 3
Object number : 1
Object number : 2
Object number : 3
```



Inline Function

- The inline functions are a C++ enhancement feature to decrease the execution time of a program.
- Functions can be instructed to compiler to make them inline so that compiler can replace function call by directly function definition.
- If function is big then, compiler can ignore the “inline” request and treat the function as normal function.



How to make function inline

- To make any function as inline, start its definitions with the keyword “inline”.

```
inline int add(int a, int b)
{
    return (a + b);
}
```



Pros/Why to use inline functions?

1. It speeds up your program by avoiding function calling overhead.
2. It save overhead of variables push/pop on the stack, when function calling happens.
3. It save overhead of return call from a function.



When to use inline functions?

Function can be made as inline as per programmer needs.
Some useful recommendations are mentioned below-

1. Use inline function when performance is needed.
2. Use inline function over macros.
3. Use inline function if function code does not have loops, iterations



Key Points

- 1.It's just a suggestion not compulsion. Compiler may or may not inline the functions you marked as inline.
- 2.It may also decide to inline functions not marked as inline at compilation time.
- 3.Inline works like a copy/paste controlled by the compiler.
- 4.Virtual methods are not supposed to be inlinable.



Program using inline function

```
#include <iostream>
using namespace std;

inline int sqr(int x)
{
    int y;
    y = x * x;
    return y;
}

int main()
{
    int a=3, b;
    b = sqr(a);
    cout <<"Ans is "<< b;
    return 0;
}
```

output: Ans is 9



Thank You..